Akoizo Original Practice - Form 2 - Answer key

Division C | Engineering Cad | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (4 points) Calculate the base volume before cutting the hole.

Authored CAD case 2. Units are mm. A rectangular base is L=70, W=32, H=10, occupying x=0..70, y=0..32, z=0..10. A cylindrical hole of radius 4 goes through the full height, centered at (x=35, y=16). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 140 g at (30,0,0) mm and 80 g at (50,0,0) mm. Do not infer any other component or coordinate.

answer: 22400 mm^3. Rectangular prism volume is length times width times height. (4 points).

Numerical target 22400; absolute tolerance 22.4. Accepted units: , mm^3.

2. (4 points) Calculate the material removed by the through hole.

Authored CAD case 2. Units are mm. A rectangular base is L=70, W=32, H=10, occupying x=0..70, y=0..32, z=0..10. A cylindrical hole of radius 4 goes through the full height, centered at (x=35, y=16). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 140 g at (30,0,0) mm and 80 g at (50,0,0) mm. Do not infer any other component or coordinate.

answer: 502.6548245743669 mm^3. Cylinder volume is pi r squared times height. (4 points).

Numerical target 502.65482; absolute tolerance 0.50265482. Accepted units: , mm^3.

3. (4 points) Calculate the remaining base volume.

Authored CAD case 2. Units are mm. A rectangular base is L=70, W=32, H=10, occupying x=0..70, y=0..32, z=0..10. A cylindrical hole of radius 4 goes through the full height, centered at (x=35, y=16). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 140 g at (30,0,0) mm and 80 g at (50,0,0) mm. Do not infer any other component or coordinate.

answer: 21897.345175425635 mm^3. (4 points).

Numerical target 21897.345; absolute tolerance 21.897345. Accepted units: , mm^3.

4. (4 points) Calculate the hole diameter.

Authored CAD case 2. Units are mm. A rectangular base is L=70, W=32, H=10, occupying x=0..70, y=0..32, z=0..10. A cylindrical hole of radius 4 goes through the full height, centered at (x=35, y=16). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 140 g at (30,0,0) mm and 80 g at (50,0,0) mm. Do not infer any other component or coordinate.

answer: 8 mm. (4 points).

Numerical target 8; absolute tolerance 0.008. Accepted units: , mm.

5. (4 points) Calculate the solid base centroid x coordinate after the centered hole is cut.

Authored CAD case 2. Units are mm. A rectangular base is L=70, W=32, H=10, occupying x=0..70, y=0..32, z=0..10. A cylindrical hole of radius 4 goes through the full height, centered at (x=35, y=16). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 140 g at (30,0,0) mm and 80 g at (50,0,0) mm. Do not infer any other component or coordinate.

answer: 35 mm. The centered cut preserves reflection symmetry. (4 points).

Numerical target 35; absolute tolerance 0.035. Accepted units: , mm.

6. (4 points) Calculate the solid base centroid y coordinate.

Authored CAD case 2. Units are mm. A rectangular base is L=70, W=32, H=10, occupying x=0..70, y=0..32, z=0..10. A cylindrical hole of radius 4 goes through the full height, centered at (x=35, y=16). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 140 g at (30,0,0) mm and 80 g at (50,0,0) mm. Do not infer any other component or coordinate.

answer: 16 mm. (4 points).

Numerical target 16; absolute tolerance 0.016. Accepted units: , mm.

7. (4 points) Calculate the solid base centroid z coordinate.

Authored CAD case 2. Units are mm. A rectangular base is L=70, W=32, H=10, occupying x=0..70, y=0..32, z=0..10. A cylindrical hole of radius 4 goes through the full height, centered at (x=35, y=16). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 140 g at (30,0,0) mm and 80 g at (50,0,0) mm. Do not infer any other component or coordinate.

answer: 5 mm. (4 points).

Numerical target 5; absolute tolerance 0.005. Accepted units: , mm.

8. (4 points) Calculate the separate two-component assembly center-of-mass x coordinate.

Authored CAD case 2. Units are mm. A rectangular base is L=70, W=32, H=10, occupying x=0..70, y=0..32, z=0..10. A cylindrical hole of radius 4 goes through the full height, centered at (x=35, y=16). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 140 g at (30,0,0) mm and 80 g at (50,0,0) mm. Do not infer any other component or coordinate.

answer: 37.27272727272727 mm. Use the mass-weighted mean, not the midpoint. (4 points).

Numerical target 37.272727; absolute tolerance 0.037272727. Accepted units: , mm.

9. (4 points) Calculate the separate assembly center-of-mass y coordinate.

Authored CAD case 2. Units are mm. A rectangular base is L=70, W=32, H=10, occupying x=0..70, y=0..32, z=0..10. A cylindrical hole of radius 4 goes through the full height, centered at (x=35, y=16). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 140 g at (30,0,0) mm and 80 g at (50,0,0) mm. Do not infer any other component or coordinate.

answer: 0 mm. (4 points).

Numerical target 0; absolute tolerance 0.001. Accepted units: , mm.

10. (4 points) A second version doubles every linear dimension of the perforated base. By what factor does its volume increase?

Authored CAD case 2. Units are mm. A rectangular base is L=70, W=32, H=10, occupying x=0..70, y=0..32, z=0..10. A cylindrical hole of radius 4 goes through the full height, centered at (x=35, y=16). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 140 g at (30,0,0) mm and 80 g at (50,0,0) mm. Do not infer any other component or coordinate.

answer: 8 . (4 points).

Numerical target 8; absolute tolerance 0.008. Accepted units: , .

11. (4 points) Convert the base length to centimetres.

Authored CAD case 2. Units are mm. A rectangular base is L=70, W=32, H=10, occupying x=0..70, y=0..32, z=0..10. A cylindrical hole of radius 4 goes through the full height, centered at (x=35, y=16). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 140 g at (30,0,0) mm and 80 g at (50,0,0) mm. Do not infer any other component or coordinate.

answer: 7 cm. (4 points).

Numerical target 7; absolute tolerance 0.007. Accepted units: , cm.

12. (4 points) Describe a feature sequence that makes the base and centered hole, including a reliable end condition for the cut.

Authored CAD case 2. Units are mm. A rectangular base is L=70, W=32, H=10, occupying x=0..70, y=0..32, z=0..10. A cylindrical hole of radius 4 goes through the full height, centered at (x=35, y=16). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 140 g at (30,0,0) mm and 80 g at (50,0,0) mm. Do not infer any other component or coordinate.

answer: Sketch a fully dimensioned rectangle on a specified plane; extrude to H; sketch the centered circle on the top face; use a remove extrude through all. Constrain the circle center and radius. Equivalent sequences that produce the same geometry are acceptable. (4 points).

13. (4 points) Why can a sketch with the correct visible shape still be under-constrained? Name a way to remove an unintended degree of freedom.

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answer: It may still translate, rotate, or change size because position, orientation or dimensions are unspecified. Add suitable dimensions/geometric constraints and anchor the intended datum; avoid redundant constraints. (4 points).

14. (4 points) Why is the arithmetic midpoint of the assembly positions generally the wrong center of mass for this case?

Authored CAD case 2. Units are mm. A rectangular base is L=70, W=32, H=10, occupying x=0..70, y=0..32, z=0..10. A cylindrical hole of radius 4 goes through the full height, centered at (x=35, y=16). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 140 g at (30,0,0) mm and 80 g at (50,0,0) mm. Do not infer any other component or coordinate.

answer: The component masses differ. Center of mass weights each coordinate by its component mass; an arithmetic midpoint assumes equal masses. (4 points).

15. (4 points) Explain how two parts can have equal volume but different center-of-mass positions.

Authored CAD case 2. Units are mm. A rectangular base is L=70, W=32, H=10, occupying x=0..70, y=0..32, z=0..10. A cylindrical hole of radius 4 goes through the full height, centered at (x=35, y=16). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 140 g at (30,0,0) mm and 80 g at (50,0,0) mm. Do not infer any other component or coordinate.

answer: Equal volume does not fix spatial distribution. Translating a part or changing shape while retaining volume can move its centroid; differing material densities can also move mass distribution. (4 points).

16. (4 points) A reported part volume is numerically correct in cm^3 but entered as mm^3. Explain the scale error.

Authored CAD case 2. Units are mm. A rectangular base is L=70, W=32, H=10, occupying x=0..70, y=0..32, z=0..10. A cylindrical hole of radius 4 goes through the full height, centered at (x=35, y=16). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 140 g at (30,0,0) mm and 80 g at (50,0,0) mm. Do not infer any other component or coordinate.

answer: One cm equals ten mm, so one cm^3 equals 1,000 mm^3. Reusing the same number with the wrong volume unit causes a factor-of-1,000 error. (4 points).

17. (4 points) Describe how to check that a hole is a cut rather than a separate overlapping solid.

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answer: Inspect the feature operation for remove/subtract, inspect a section through the hole, check that the passage is empty and verify that the remaining volume equals prism volume minus cylinder volume. (4 points).

18. (4 points) Explain why matching the part volume alone does not prove that an assembly is correct.

Authored CAD case 2. Units are mm. A rectangular base is L=70, W=32, H=10, occupying x=0..70, y=0..32, z=0..10. A cylindrical hole of radius 4 goes through the full height, centered at (x=35, y=16). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 140 g at (30,0,0) mm and 80 g at (50,0,0) mm. Do not infer any other component or coordinate.

answer: Incorrectly positioned or oriented components can retain their individual volumes. Check specified coordinates, mates, materials and assembly center of mass as well as individual geometry. (4 points).